Problem 5.13.16

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Question

Question: Let k be an integer such that the triangle with vertices

$$\mathbf{A} = \begin{pmatrix} k \\ -3k \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ k \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -k \\ 2 \end{pmatrix}$$

has area 28 sq. units. Find the orthocenter of the triangle.

Solution

$$\text{Area} = \frac{1}{2}\left|\mathbf{M}\right| = 28 \tag{2.1}$$

$$\text{where} \quad \mathbf{M} = \begin{pmatrix} \mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} \end{pmatrix} = \begin{pmatrix} 5-k & -2k \\ 4k & 2+3k \end{pmatrix} \tag{2.2}$$

$$\det(\mathbf{M}) = (5-k)(2+3k) - (-2k)(4k) = 10 + 13k + 5k^2 \tag{2.3}$$

$$10 + 13k + 5k^2 = \pm 56 \tag{2.4}$$

After solving,

$$k = 2 \tag{2.5}$$

Solution

$$\text{For } k = 2, \quad \mathbf{A} = \begin{pmatrix} 2 \\ -6 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} \tag{2.6}$$

The equation of the altitude through $\mathbf{A}$ is:

$$\mathbf{x}^T \mathbf{n} = 1 \tag{2.7}$$

This altitude passes through $\mathbf{A}$, so

$$\mathbf{A}^T \mathbf{n} = 1 \tag{2.8}$$

$$\begin{pmatrix} 2 & -6 \end{pmatrix} \mathbf{n} = 1 \tag{2.9}$$

and the direction vector $\mathbf{n}$ of the altitude is perpendicular to $\mathbf{B} - \mathbf{C}$, so

$$(\mathbf{B} - \mathbf{C})^T \mathbf{n} = 0 \tag{2.10}$$

$$\begin{pmatrix} 7 & 0 \end{pmatrix} \mathbf{n} = 0 \tag{2.11}$$

Solution

The augmented matrix is:

$$\left(\begin{array}{cc|c} 7 & 0 & 0 \\ 2 & -6 & 1 \end{array}\right) \tag{2.12}$$

Row reduction steps:
First, $R_2 \to R_2 - \frac{2}{7}R_1$:

$$\left(\begin{array}{cc|c} 7 & 0 & 0 \\ 0 & -6 & 1 \end{array}\right) \tag{2.13}$$

$$\mathbf{n} = \begin{pmatrix} 0 \\ -\frac{1}{6} \end{pmatrix} \tag{2.14}$$

The equation of the altitude through $\mathbf{B}$ is:

$$\mathbf{x}^T\mathbf{n} = 1 \tag{2.15}$$

Solution

This altitude passes through $\mathbf{B}$, so

$$\mathbf{B}^T\mathbf{n} = 1 \tag{2.16}$$

$$\begin{pmatrix} 5 & 2 \end{pmatrix}\mathbf{n} = 1 \tag{2.17}$$

and the direction vector $\mathbf{n}$ of the altitude is perpendicular to $\mathbf{A} - \mathbf{C}$, so

$$(\mathbf{A} - \mathbf{C})^T\mathbf{n} = 0 \tag{2.18}$$

$$\begin{pmatrix} 4 & -8 \end{pmatrix}\mathbf{n} = 0 \tag{2.19}$$

The augmented matrix is:

$$\left(\begin{array}{cc|c} 5 & 2 & 1 \\ 4 & -8 & 0 \end{array}\right) \tag{2.20}$$

Solution

Row reduction steps: First, $R_2 \to R_2 - \frac{4}{5}R_1$:

$$\left(\begin{array}{cc|c} 5 & 2 & 1 \\ 0 & -\frac{48}{5} & -\frac{4}{5} \end{array}\right) \tag{2.21}$$

Next, $R_2 \to -\frac{5}{48}R_2$:

$$\left(\begin{array}{cc|c} 5 & 2 & 1 \\ 0 & 1 & \frac{1}{12} \end{array}\right) \tag{2.22}$$

Back substitute: $R_1 \to R_1 - 2R_2$

$$\left(\begin{array}{cc|c} 5 & 0 & \frac{5}{6} \\ 0 & 1 & \frac{1}{12} \end{array}\right) \tag{2.23}$$

$$\mathbf{n} = \begin{pmatrix} \frac{1}{6} \\ \frac{1}{12} \end{pmatrix} \tag{2.24}$$

Solution

First altitude:

$$\begin{pmatrix} 0 & -\frac{1}{6} \end{pmatrix} \mathbf{x} = 1 \tag{2.25}$$

Second altitude:

$$\begin{pmatrix} \frac{1}{6} & \frac{1}{12} \end{pmatrix} \mathbf{x} = 1 \tag{2.26}$$

Combining both equations:

$$\begin{pmatrix} 0 & -\frac{1}{6} \\ \frac{1}{6} & \frac{1}{12} \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{2.27}$$

Writing the augmented matrix:

$$\left(\begin{array}{cc|c} \frac{1}{6} & \frac{1}{12} & 1 \\ 0 & -\frac{1}{6} & 1 \end{array}\right) \tag{2.28}$$

Solution

Perform the row operation $R_1 \to R_1 + \frac{1}{2}R_2$:

$$\left(\begin{array}{cc|c} \frac{1}{6} & 0 & \frac{3}{2} \\ 0 & -\frac{1}{6} & 1 \end{array}\right) \tag{2.29}$$

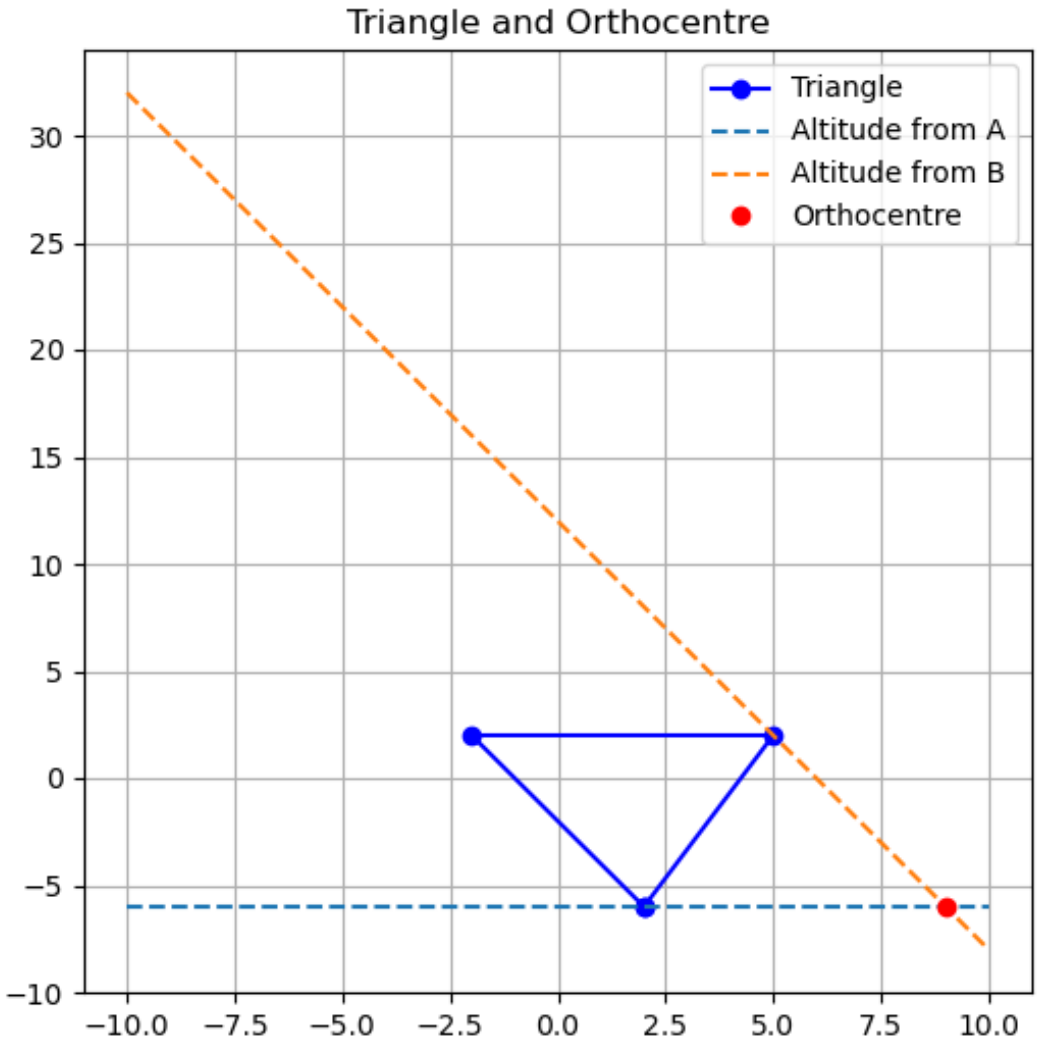
Therefore, the solution vector is:

$$\mathbf{x} = \begin{pmatrix} 9 \\ -6 \end{pmatrix} \tag{2.30}$$

Answer: The orthocentre of the triangle is

$$\boxed{\mathbf{x} = \begin{pmatrix} 9 \\ -6 \end{pmatrix}}$$

Graph


Triangle and Orthocentre
Triangle
Altitude from A
Altitude from B
Orthocentre
30
25
20
15
10
5
0
−5
−10
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0

C Code

```c
#include "matfun.h"

void line_equation(const double P[2], const double Q[2], const
    double R[2], double *a, double *b, double *c) {
    double dx = R[0] - Q[0];
    double dy = R[1] - Q[1];
    *a = -dy;
    *b = dx;
    *c = (*a)*P[0] + (*b)*P[1];
}

void solve_2x2(double a1, double b1, double c1, double a2, double
     b2, double c2, double sol[2]) {
    double det = a1*b2 - a2*b1;
    sol[0] = (b2*c1 - b1*c2) / det;
    sol[1] = (a1*c2 - a2*c1) / det;
}
```

C Code

```c
void compute_orthocentre(const double A[2], const double B[2],
    const double C[2], double O[2]) {
    double a1, b1, c1, a2, b2, c2;
    line_equation(A, B, C, &a1, &b1, &c1);
    line_equation(B, A, C, &a2, &b2, &c2);
    solve_2x2(a1, b1, c1, a2, b2, c2, O);
}
```

Python + C Code

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

lib = ctypes.CDLL('./matfun.so')

arr_type = ctypes.c_double * 2
A = arr_type(2, -6)
B = arr_type(5, 2)
C = arr_type(-2, 2)
O = arr_type()

lib.compute_orthocentre.argtypes = [arr_type, arr_type, arr_type,
     arr_type]
lib.line_equation.argtypes = [arr_type, arr_type, arr_type,
    ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
    c_double), ctypes.POINTER(ctypes.c_double)]
```

Python + C Code

```
lib.solve_2x2.argtypes = [ctypes.c_double, ctypes.c_double,
    ctypes.c_double, ctypes.c_double, ctypes.c_double, ctypes.
    c_double, arr_type]

# Call orthocentre computation
lib.compute_orthocentre(A, B, C, O)
print("Orthocentre:", list(O))

# Arrays to hold line coefficients
a1, b1, c1 = ctypes.c_double(), ctypes.c_double(), ctypes.
    c_double()
a2, b2, c2 = ctypes.c_double(), ctypes.c_double(), ctypes.
    c_double()
a3, b3, c3 = ctypes.c_double(), ctypes.c_double(), ctypes.
    c_double()
```

Python + C Code

```
# Compute altitude lines via C functions for all three vertices
lib.line_equation(A, B, C, ctypes.byref(a1), ctypes.byref(b1),
    ctypes.byref(c1)) # Altitude from A
lib.line_equation(B, A, C, ctypes.byref(a2), ctypes.byref(b2),
    ctypes.byref(c2)) # Altitude from B
lib.line_equation(C, A, B, ctypes.byref(a3), ctypes.byref(b3),
    ctypes.byref(c3)) # Altitude from C

# Convert points to numpy for plotting
A_np = np.array([2, -6])
B_np = np.array([5, 2])
C_np = np.array([-2, 2])
O_np = np.array([O[0], O[1]])

fig, ax = plt.subplots(figsize=(6,6))
ax.plot([A_np[0], B_np[0], C_np[0], A_np[0]], [A_np[1], B_np[1],
    C_np[1], A_np[1]], 'bo-', label='Triangle')
```

Python + C Code

```
def plot_line(a, b, c, ax, color, label):
    if abs(b) > 1e-8:
        x = np.linspace(-10, 20, 300)
        y = (c - a*x)/b
    else:
        x = np.full(300, c/a)
        y = np.linspace(-10, 20, 300)
    ax.plot(x, y, color=color, linestyle='--', label=label)

plot_line(a1.value, b1.value, c1.value, ax, 'r', 'Altitude from A
    ')
plot_line(a2.value, b2.value, c2.value, ax, 'g', 'Altitude from B
    ')
plot_line(a3.value, b3.value, c3.value, ax, 'b', 'Altitude from C
    ')
```

Python + C Code

```
ax.plot(O_np[0], O_np[1], 'ro', label='Orthocentre')

ax.legend()
ax.grid(True)
ax.set_title('Triangle, Altitudes, and Orthocentre (using .so
    functions)')
plt.show()
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([2, -6])
B = np.array([5, 2])
C = np.array([-2, 2])

def altitude_through_point(P, Q, R):
    """
    Returns coefficients a,b,c of the altitude through P
    which is perpendicular to QR, i.e. a*x + b*y = c
    """
    # Direction vector of QR
    d = Q - R
    # Normal vector for the altitude (perpendicular)
    n = np.array([-d[1], d[0]])
    c = n[0]*P[0] + n[1]*P[1]
    return n[0], n[1], c
```

Python Code

```
def intersect_lines(a1, b1, c1, a2, b2, c2):
    """
    Finds the intersection point of two lines: a1*x+b1*y=c1 and
        a2*x+b2*y=c2
    """
    A = np.array([[a1, b1], [a2, b2]])
    b = np.array([c1, c2])
    return np.linalg.solve(A, b)

a1, b1, c1 = altitude_through_point(A, B, C) # through A, perp to
     BC
a2, b2, c2 = altitude_through_point(B, A, C) # through B, perp to
     AC

O = intersect_lines(a1, b1, c1, a2, b2, c2)

print("Orthocentre:", O)
```

Python Code

```
# Plotting
plt.figure(figsize=(6,6))
plt.plot([A[0], B[0], C[0], A[0]], [A[1], B[1], C[1], A[1]], 'bo-
    ', label='Triangle')

# Plot altitudes
def plot_altitude(P, n, c, label):
    # Generate points along the altitude for plot
    t = np.linspace(-10, 10, 100)
    if n[1] != 0:
        x = t
        y = (c - n[0]*x)/n[1]
    else:
        x = np.ones_like(t)*(c/n[0])
        y = t
    plt.plot(x, y, '--', label=label)
```

Python Code

```
plot_altitude(A, np.array([a1, b1]), c1, 'Altitude from A')
plot_altitude(B, np.array([a2, b2]), c2, 'Altitude from B')

plt.plot(O[0], O[1], 'ro', label='Orthocentre')
plt.legend()
plt.grid(True)
plt.title('Triangle and Orthocentre')
plt.savefig("fig1.png")
plt.show()
```